BEING QUASI-SEPARATED IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one Morphisms of Spaces lemma, one Topologies on Spaces lemma and the same-chapter quasi-compactness descent, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Base change of (quasi-)separated morphisms. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-separated. Proves the direct direction.
- Base change of fpqc coverings. Stacks Project, Topologies on Spaces, Lemma spaces-topologies-lemma-fpqc. Shows the base-changed family is again an fpqc covering.
- fpqc descent of quasi-compactness. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-quasi-compact. Descends quasi-compactness of the diagonal.

Source reference. Original reference 03KL: sources/stacks-chow/tags.txt, source lines 4681–4681. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 03YH: sources/stacks-chow/tags.txt, source lines 5184–5184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041L: sources/stacks-chow/tags.txt, source lines 5296–5296. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. The property $\mathcal{P}(f) =$"f is quasi-separated" is fpqc local on the base on algebraic spaces over S.*

Proof. A base change of a quasi-separated morphism is quasi-separated, see Morphisms of Spaces, Lemma 03KL. Hence the direct implication in Definition 03YH.

Let $\{Y_i \to Y\}_{i \in I}$ be an fpqc covering of algebraic spaces over S. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume each base change $X_i := Y_i \times_Y X \to Y_i$ is quasi-separated. This means that each of the morphisms

$$\Delta_i : X_i \longrightarrow X_i \times_{Y_i} X_i = Y_i \times_Y (X \times_Y X)$$

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is quasi-compact. The base change of a fpqc covering is an fpqc covering, see Topologies on Spaces, Lemma 03MR hence $\{Y_i \times_Y (X \times_Y X) \to X \times_Y X\}$ is an fpqc covering of algebraic spaces. Moreover, each Δ_i is the base change of the morphism $\Delta : X \to X \times_Y X$. Hence it follows from Lemma 041L that Δ is quasi-compact, i.e., f is quasi-separated. $\square$